KONFERENCJA RADIOKOMUNIKACJI
I TELEINFORMATYKI
KRiT 2025

HD-dSEQUA: Baza zróżnicowanych sekwencji HD do obiektywnej oceny jakości wraz z indykatorami

HD-dSEQUA: High Definition Diverse Video Sequence Dataset for Quality Assessment with Indicators

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Streszczenie: Wzrost popularności treści wideo w Internecie oraz postęp w dziedzinie automatycznego przetwarzania obrazu powodują rosnące zapotrzebowanie na narzędzia służące do porównywania algorytmów kompresji. W odpowiedzi opracowano autorską bazę sekwencji wideo, zawierającą 131 materiałów o różnorodnych charakterystykach przestrzennych, czasowych i semantycznych. W artykule przedstawiono strukturę bazy, kryteria doboru sekwencji oraz metryki jakości wideo pozwalające na szybkie porównanie parametrów materiału wideo.

Abstract: The growing popularity of video content on the Internet and the development of automatic image processing systems have created a need for appropriate testing tools to enable comparison of compression algorithms. In response, a video author dataset has been developed that contains 131 sequences with various spatial, temporal, and semantic characteristics. The paper presents the structure of the data set, the sequence selection criteria, and video quality metrics that allow for quick comparison of video parameters.

Słowa kluczowe: baza sekwencji, kompresja wideo, metryki jakości wideo, ocena jakości, przetwarzanie obrazu

Keywords: video processing, quality assessment, video dataset, video compression, video quality metrics

1. INTRODUCTION

According to [3], global IP traffic reached 120 zettabytes in 2023 and is projected to increase to 181 zettabytes by 2025, with video content accounting for more than half of the total data volume. This substantial share highlights the exponential growth of video usage across diverse domains – such as entertainment, surveillance, education, healthcare, artificial intelligence, and more – which in turn underscores the need for diverse video datasets to support the evaluation and comparison of video processing systems.

Such systems consist of video encoding techniques, quality metrics, and machine learning (ML) models. Selecting a proper training database for an ML model is crucial because it directly influences the model's performance, generalizability, and robustness. A well-designed data set ensures that the model learns relevant patterns and features, avoids overfitting to narrow content, and performs reliably in diverse real-world scenarios.

A wide variety of video quality databases are available for research and evaluation purposes. Notable examples include

- **YouTube UGC** [10] – a data set comprising 1,500 user-generated videos, each 20 seconds long, with resolutions ranging from 360p to 1080p. The videos span diverse categories such as gaming, sports, and more, and have been evaluated using no-reference quality metrics.
- **BVI-HD** [8] – a collection consisting of 32 reference videos and 384 corresponding distorted versions, along with associated Differential Mean Opinion Scores (DMOS). This data set has been used to assess subjective video quality under HEVC compression and to benchmark the performance of various reference quality metrics.
- **KoNViD-150k** [4] — a dataset comprising 153,841 publicly available video clips, carefully selected from various sources to ensure a diverse range of content and visual characteristics. Each video includes five quality ratings, with a subset of 1,596 videos receiving up to 89 ratings each.

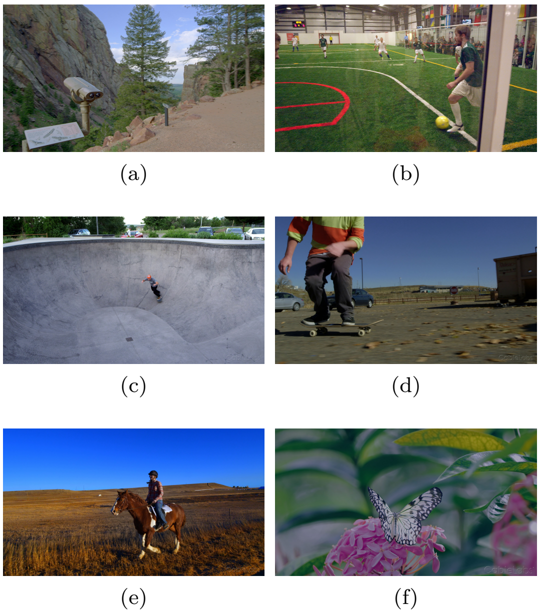
The main limitation of these databases, as well as many others, is the absence or very limited availability of videos in raw format. Raw video data are particularly valuable as they enable the introduction of various distortions – such as compression – into the source material, allowing for comprehensive testing of video quality under various conditions. One of the most widely used [1, 2, 9] full reference metrics for this purpose is the Video Multimethod Assessment Fusion (VMAF), developed by Netflix [5].

2. VIDEOS

The primary goal when selecting video sequences for HD-dSEQUA was to include as diversified video content as possible, not limited to animated or professionally generated content but also encompassing a wide variety of real-world footage. This approach was intended to ensure broad coverage of visual characteristics such as motion and spatial complexity, texture variation, and lighting conditions, making the data set robust and representative of different visual scenarios.

2.1 Sources

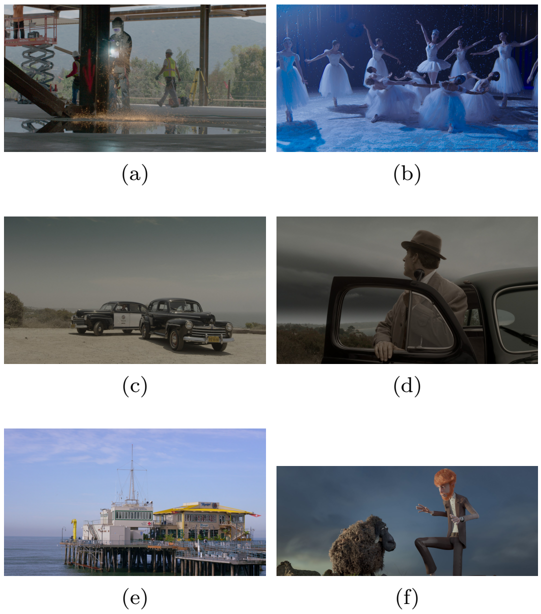
The video sequences were collected from several publicly available datasets and video content sources. A significant portion of the material originates from CableLabs, which offers a diverse range of 1080p videos, including both user-generated content (UGC) and professional productions. Representative sequences include Ancient Thought, Eldorado (Fig. 1a), Indoor Soccer (Fig. 1b), Moment of Intensity (Fig. 1c), Skateboarding (Fig. 1d), Unspoken Friend (Fig. 1e), and Wipe HDR (Fig. 1f). Among these, Moment of Intensity and Wipe HDR are high frame rate sequences recorded at 60 FPS, while the remaining videos are captured at 24 FPS.



(a) (b) (c) (d) (e) (f)

Figure 1: Frame examples of the CableLabs sequences.

Additional sequences were sourced from Netflix, featuring a wide variety of professionally produced content at different resolutions and frame rates. Notable 1080p sequences recorded at 60 FPS include Sparks (Fig. 2a), ballet (Fig. 2b), beach, cave, cliff, driving, inside, leaving, police (Fig. 2c), stairs, talkinghead, and telephone (Fig. 2d). The Chimera sequences, mostly at 1080p and 30 FPS, include several detailed scenes such as Aerial, Driving POV, Pier Seaside (Fig. 2e), Roller Coaster Passengers, and Toddler Fountain. The Cosmos Laundromat collection (Fig. 2f) consists of multiple 1920×804 sequences at 24 FPS, with natural and synthetic content. Additionally, 4K 60 FPS sequences include El Fuente scenes like Boat and Food Market, as well as Chimera Dancers.



(a) (b) (c) (d) (e) (f)

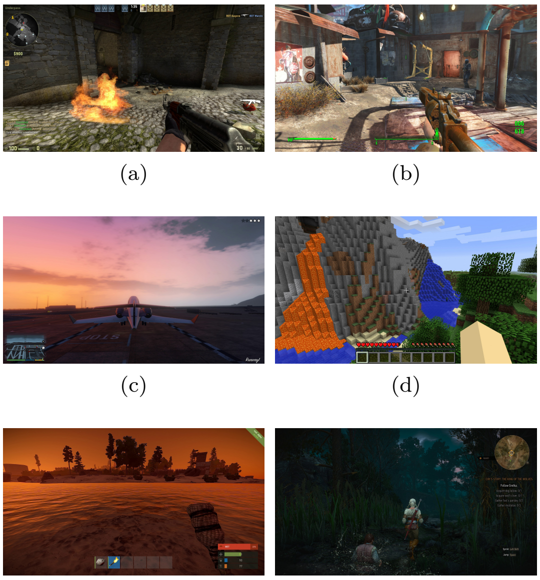
Figure 2: Frame examples of the Netflix sequences.

The SJTU Media Lab contributed a comprehensive set of sequences depicting both urban and natural environments, all captured in 1080p resolution at 25 FPS. Representative videos include Rush Hour (Fig. 3a), Campfire Party, and Tall Buildings (Fig. 3c), along with additional scenes such as Bund Nightscape (Fig. 3b), Library, Tree Shade, Construction Field, Fountains, Marathon, Residential Building, Runners (Fig. 3d), Scarf, Traffic Flow, Traffic and Building (Fig. 3e), and Wood (Fig. 3f).

(a) (b) (c) (d) (e) (f)

Figure 3: Frame examples of the SJTU Media Lab sequences.

In addition, dynamic gameplay and streaming content were incorporated from the Xiph.org Video Test Media and Ultra Video Group datasets, as well as a wide range of Twitch-captured streams, facilitating the analysis of real-time rendering and interactive media. The Twitch collection features 1080p sequences at 60 FPS across numerous popular game titles, including CS:GO (Fig. 4a), DOTA2, Euro Truck Simulator 2, Fallout 4 (Fig. 4b), GTA V (Fig. 4c), Hearthstone, Minecraft (Fig. 4d), Rust (Fig. 4e), Starcraft and The Witcher 3 (Fig. 4f). This collection covers a range of motion complexity, visual patterns, camera pans, and gameplay perspectives.



(a) (b) (c) (d) (e) (f)

Figure 4: Frame examples of the gameplay sequences.

2.2 Metadata and Statistics

The videos in the data set are stored in the Y4M format, which preserves the quality of uncompressed raw video. This format does not introduce compression artifacts, ensuring an accurate representation of the source content for processing. The use of Y4M guarantees consistency across the data set and is particularly well suited for research and benchmarking tasks in video quality assessment, encoding, and streaming performance evaluation.

Of the 131 videos in the dataset, 117 are in Full HD resolution (1920×1080), 11 have a resolution of 1920×804, and 3 are available in Ultra HD (3840×2160). In terms of frame rates, 39 videos are recorded at 24 FPS, 16 at 25 FPS, 5 at 30 FPS, and the remaining 71 at 60 FPS.

The durations of the video sequences are presented in a histogram in Figure 5. Most of the videos have durations between 10 and 20 seconds, with the highest concentration around 13 to 16 seconds – where the number of sequences peaks at 48. The interquartile range, from the 25th percentile at 11.93 seconds to the 75th percentile at 15 seconds, confirms that half of the videos are tightly clustered within this short duration window. The median duration is 14 seconds, while the mean is slightly higher at 17.12 seconds. There are also a noticeable number of shorter videos between 7 and 13 seconds, and very few sequences shorter than 7 seconds. During 20 seconds, the number of sequences drops dramatically. A distinct outlier exists at the 60-second mark, which aligns with the maximum value of 60.05 seconds and contributes to the elevated mean. The standard deviation of 12.80 seconds reflects the overall spread in video durations. In summary, the majority of the videos are short and fairly consistent in length, with a small number of much longer videos that skew the distribution.

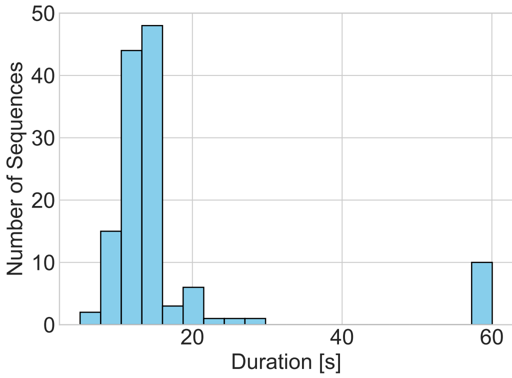


Figure 5: Distribution of video sequence durations.

3. VIDEO QUALITY INDICATORS

The video sequences were assessed using 10 no-reference Video Quality Indicators (VQIs) developed at AGH [7, 6]. **Spatial Activity (SA)** captures the amount of spatial detail within each frame, while **Temporal Activity (TA)** reflects the intensity of movement [7]. **Blockiness (BLN)** detects compression artifacts, and **Blockloss (BLL)** measures the impact of lost data blocks on visual quality [7]. **Blur (BLU)** assesses the loss of sharpness, often due to compression, removing high-frequency details [7]. **Exposure (EXP)** and **Contrast (CON)** evaluate the lighting balance and the variation of brightness, respectively [7]. **Interlace (INT)** identifies interlacing artifacts, and **Noise (NOI)** captures random brightness or color fluctuations [7]. Finally, **UGC** distinguishes user-generated content from professional video [6].

All of these indicators were computed using the agh-vqis Python package [6], which can be easily installed via pip. The metrics are lightweight and efficient, making them well-suited for real-time video quality assessment.

Figure 6 presents the average values of SA and TA for various video sequences. Higher values indicate greater visual detail or more motion within a video. Several sequences exhibit notably high activity levels, such as sequence 107, which features dense tree coverage combined with rapid camera panning, and sequence 41, showcasing Fallout gameplay with intense detail and fast camera movements, including elements like trees, sand, fire, and rain. In contrast, sequences 101 and 104 show low SA and TA values, corresponding to a blurry recording of a caterpillar moving slowly along a branch.

Table 1 summarizes the descriptive statistics of the VQIs computed on the entire video sequence database.

The BLN and BLL metrics show relatively low mean values (0.99 and 1.50, respectively), although BLL displays high variability, as indicated by its large standard deviation (2.47) and maximum value (14.97). This suggests that while most sequences exhibit minor block loss, a subset suffers from severe degradation.

The BLU metric has a mean of 4.575 and a wide range, from 1.33 to 12.59, reflecting a significant variation in sharpness in the data set.

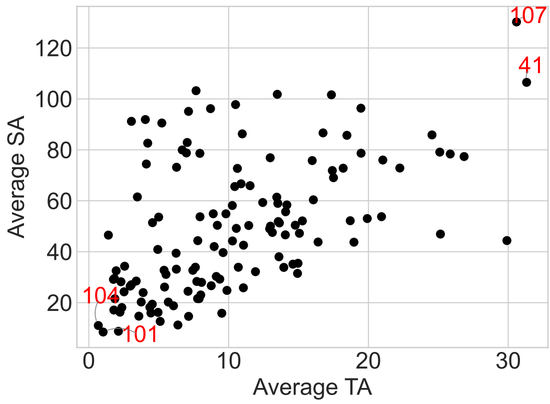


Figure 6: Average SA vs Average TA for each sequence.

EXP and CON both show moderate variability. EXP ranges from 64.77 to 143.34, with a mean of 111.32 and a standard deviation of 16.96, suggesting a well-lit data set, but with some under- and overexposed content. CON follows a similar pattern, ranging from 7.93 to 73.66.

The INT artifact is largely absent in the dataset, as reflected by its mean, median, and lower quartiles, all being zero. The maximum value of 0.03 indicates only rare occurrences of interlacing. The INT column was excluded from Table 1, as its values were almost entirely zero.

NOI shows a moderate mean of 1.44 and a substantial maximum of 6.52, indicating that some sequences are affected by significant noise, likely due to sensor limitations in UGC.

Lastly, the UGC classifier shows a bimodal distribution, with a median and 75th percentile of 1.00 and a mean of 0.67. This suggests that a large portion of the data set consists of UGC, but not all.

Table 1: Statistics of VQIs computed for the database.

	BLN	*BLL*	*BLU*	*EXP*	*CON*	*NOI*	*UGC*
mean	0.99	1.50	4.58	111.32	35.48	1.44	0.67
std	0.06	2.47	1.85	16.96	11.65	1.25	0.46
min	0.50	0.00	1.33	64.77	7.93	0.00	0.00
25%	0.99	0.36	3.40	100.51	26.87	0.65	0.00
50%	1.00	0.74	4.23	115.29	34.91	0.90	1.00
75%	1.00	1.68	5.25	125.58	44.21	1.70	1.00
max	1.09	14.97	12.59	143.34	73.66	6.52	1.00

A comprehensive table containing sequence IDs, names, and associated Video Quality Indicator values is provided in the CSV file available at:
https://qoe.agh.edu.pl/hd-dsequa.
All video sequences can also be accessed from the same location.

4. CONCLUSIONS AND FUTURE WORK

We introduced a diverse dataset of 131 raw video sequences designed for objective quality assessment and video compression research. Each sequence includes key no-reference quality metrics, enabling quick and consistent evaluation. The data set fills a gap in the availability of high-quality, diverse raw content suitable for benchmarking modern video systems. Future work will focus on expanding the dataset, adding distortion variants, and integrating subjective assessments to support broader research applications.

5. ACKNOWLEDGMENTS

The work was funded by the Ministry of Science and Higher Education as part of the subsidy for the Faculty of Computer Science, Electronics, and Telecommunications at AGH University of Krakow.

The presented research results come from a research and development project entitled „Technology for Secure and Reliable Delivery of Professional Audio/Video Contribution Live Transmissions with the Lowest Possible Latency." The project was carried out under Submeasure 1.1.1 of the Intelligent Development Operational Program 2014–2020 and co-financed from the Polish National Center for Research and Development. It was implemented as part of the National Center for Research and Development's Fast Path competition.

REFERENCES

[1] Amirpour Hadi et al. 2022. Between Two and Six? Towards Correct Estimation of JND Step Sizes for VMAF-based Bitrate Laddering. 2022 14th International Conference on Quality of Multimedia Experience (QoMEX), 1–4.

[2] Mercat Alexandre et al. 2021. Comparative Rate-Distortion-Complexity Analysis of VVC and HEVC Video Codecs. IEEE Access, 9: 67813–67828.

[3] Shirvani Moghaddam Shahriar. 2024. The Past, Present, and Future of the Internet: A Statistical, Technical, and Functional Comparison of Wired/Wireless Fixed/Mobile Internet. Electronics, 13(10): 1986.

[4] Götz-Hahn Franz et al. 2021. KonVid-150k: A Dataset for No-Reference Video Quality Assessment of Videos in-the-Wild. IEEE Access, 9: 72139–72160.

[5] Li Zhi et al. 2016. Toward a Practical Perceptual Video Quality Metric. The Netflix Tech Blog.

[6] Leszczuk Mikołaj et al. 2023. "In the Wild" Video Content as a Special Case of User Generated Content and a System for Its Recognition. Sensors, 23(4): 1769.

[7] Leszczuk Mikołaj et al. 2023. Objective Video Quality Assessment and Ground Truth Coordinates for Automatic License Plate Recognition. Electronics, 12(23): 4721.

[8] Zhang Fan et al. 2018. BVI-HD: A Video Quality Database for HEVC Compressed and Texture Synthesized Content. IEEE Transactions on Multimedia, 20(10): 2620–2630.

[9] Uddin Syed et al. 2022. Preliminary Study on Video Codec Optimization Using VMAF. Intelligent Information and Database Systems, 469–480.

[10] Wang Yilin et al. 2019. YouTube UGC Dataset for Video Compression Research. 2019 IEEE 21st International Workshop on Multimedia Signal Processing (MMSP), 1–5.